

NATIONAL ECONOMICS UNIVERSITY



Modeling COVID-Period Dynamics in Anxiety Search Behavior: An ARIMA, SARIMA, and SARIMAX Intervention Analysis Across Six Anglophone Countries

Course: Time Series

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Abstract

This study examines the time-series dynamics of anxiety-related Google search behavior across six Anglophone countries: Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States. Using monthly Google Trends topic-level data from January 2004 to December 2025, the paper applies ARIMA, SARIMA, and SARIMAX intervention models to analyse autoregressive dependence, monthly seasonality, and COVID-period structural changes in online mental-health search behavior. The main empirical focus is anxiety search interest, while naive and seasonal-naive models are included as benchmark references rather than as the central methodological contribution.

The analysis shows that anxiety-related search behavior is strongly persistent and seasonal across countries. ARIMA provides a non-seasonal baseline, while SARIMA captures recurring monthly patterns and improves short-horizon forecasting performance. However, over the post-COVID test period, naive and seasonal-naive benchmarks outperform ARIMA, SARIMA, and SARIMAX in most countries, indicating that anxiety search behavior became highly stable and seasonally predictable after the COVID period. This benchmark dominance does not invalidate the ARIMA/SARIMA/SARIMAX framework; instead, it clarifies the nature of post-COVID search dynamics.

SARIMAX intervention models are used to estimate the effects of early-COVID shocks, COVID-period trends, and post-COVID adjustments. The results do not provide robust evidence of a uniform positive COVID-period increase in anxiety searches after controlling for autoregressive and seasonal dynamics. Significant intervention effects are country-specific and mostly point toward post-COVID decline or normalization rather than sustained COVID-driven increases. Counterfactual SARIMA analysis further shows that observed anxiety search interest generally does not exceed the no-COVID baseline projected from pre-COVID dynamics, especially when bounded-scale robustness checks are considered.

Overall, the findings suggest that COVID-period anxiety search behavior is better understood as heterogeneous, persistent, seasonal, and model-sensitive rather than as a simple cross-national increase. The study demonstrates the value of ARIMA, SARIMA, and SARIMAX not merely as forecasting tools, but as structured methods for decomposing time-series dependence, seasonality, and intervention effects in digital mental-health indicators.

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1 Introduction

1.1 Research Context

Mental health became a major public-health and social concern during the COVID-19 period. Lockdowns, health uncertainty, economic disruption, isolation, and changes in work and education affected how individuals experienced and expressed psychological distress. Alongside survey and clinical indicators, online search behavior provides a useful source of real-time behavioral evidence because individuals often search for symptoms, coping strategies, and mental-health information.

This study uses Google Trends data to examine anxiety-related search behavior across six Anglophone countries: Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States. Google Trends does not measure clinical prevalence, but it captures relative public search interest over time. Therefore, anxiety-related search interest is treated as a digital proxy for public attention to anxiety, not as a direct measure of diagnosed anxiety disorders.

1.2 Research Problem

A simple approach would ask whether COVID-19 increased anxiety-related searches. However, this question is incomplete because anxiety search behavior may already contain persistence, long-term trends, and monthly seasonality before COVID-19. If these time-series properties are ignored, post-2020 changes may be incorrectly interpreted as COVID effects.

This paper therefore examines how anxiety search behavior changed after accounting for autoregressive dependence, seasonality, and intervention effects. ARIMA is used to model non-seasonal dependence, SARIMA captures monthly seasonal dynamics, and SARIMAX estimates COVID-period and post-COVID intervention effects. Naive and seasonal-naive models are included only as benchmark references for forecast evaluation.

1.3 Research Objectives

The objective of this study is to apply ARIMA, SARIMA, and SARIMAX models to examine the dynamics of anxiety-related Google search behavior from January 2004 to December 2025. Specifically, the paper aims to:

1. identify long-term trend, persistence, and monthly seasonality in anxiety-related searches;
2. use stationarity tests, ACF plots, and PACF plots to justify ARIMA and SARIMA candidate orders;
3. compare ARIMA, SARIMA, and SARIMAX models with naive and seasonal-naive benchmarks;
4. estimate SARIMAX intervention effects for the COVID and post-COVID periods;

5. construct SARIMA counterfactual forecasts to compare observed search behavior with a no-COVID baseline.

1.4 Research Questions

This paper is guided by five research questions:

1. Do anxiety-related Google searches exhibit autoregressive dependence and monthly seasonality?
2. How do ARIMA and SARIMA characterize non-seasonal and seasonal search dynamics?
3. Do SARIMAX intervention models detect significant COVID-period or post-COVID level and trend shifts?
4. Do ARIMA, SARIMA, or SARIMAX improve forecast performance relative to naive benchmarks?
5. How do anxiety search dynamics differ across the six countries?

1.5 Main Contributions

This paper makes three contributions. First, it applies a structured ARIMA–SARIMA–SARIMAX framework to long monthly Google Trends data across six comparable countries. Second, it separates forecast accuracy from structural interpretation. Although naive and seasonal-naive benchmarks perform strongly in the post-COVID test period, ARIMA, SARIMA, and SARIMAX remain important because they model dependence, seasonality, and intervention effects explicitly. Third, it provides a cautious interpretation of the COVID effect. The results do not support a uniform positive COVID-period increase in anxiety searches; instead, the evidence suggests country-specific and mostly post-COVID normalization or decline.

1.6 Report Structure

The report proceeds as follows. Section ?? reviews relevant background on Google Trends, mental-health search behavior, and time-series intervention modeling. Section 2 describes the data and preprocessing procedure. Section 3 presents exploratory time-series analysis. Section ?? explains the ARIMA, SARIMA, SARIMAX, benchmark, diagnostic, and counterfactual methods. Sections ??–7 report the empirical results, forecast evaluation, intervention findings, and cross-country comparison. Sections ??–?? discuss implications, limitations, and conclusions.

2 Data Understanding and Preprocessing

2.1 Dataset Overview

This study uses monthly Google Trends topic-level search indices to examine mental-health-related search behavior across six English-speaking countries. The countries included in the analysis are Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States. The raw sample covers January 2004 to December 2025, giving 264 monthly observations for each country. The full raw panel therefore contains:

$$6 \text{ countries} \times 264 \text{ months} = 1,584 \text{ observations.} \quad (1)$$

The main target variable in this study is the Google Trends topic-level index for anxiety. Four additional mental-health-related search variables are retained for exploratory analysis and robustness comparison: depression, stress, insomnia, and mental disorder. These variables are not interpreted as direct clinical prevalence measures. Instead, they are treated as normalized indicators of relative public search interest over time.

Table 1: Raw Dataset Overview

Item	Description
Data source	Google Trends topic-level search index
Countries	Australia, Canada, Ireland, New Zealand, UK, US
Frequency	Monthly
Raw period	January 2004 to December 2025
Months per country	264
Total raw observations	1,584
Main modeling variable	Anxiety
Additional variables	Depression, stress, insomnia, mental disorder
Raw file structure	Six separate country-level CSV files

The raw data are stored as six country-level CSV files. Each file contains a monthly timestamp and five Google Trends topic-level variables. Before analysis, all country files are loaded, standardized, and combined into one balanced country-month panel.

2.2 Column Standardization

The raw Google Trends files use descriptive topic names. For consistency across countries and later modeling stages, these names are standardized into concise variable names. The raw column **Time** is converted into the monthly date index. The topic names are renamed as follows.

Table 2: Variable Name Standardization

Raw column name	Standardized name
Time	date
Anxiety	anxiety
Major depressive disorder	depression
Psychological Stress	stress
Insomnia	insomnia
Mental disorder	mental_disorder

After renaming, each country-level dataset is converted into the following standard structure:

`date, country, anxiety, depression, stress, insomnia, mental_disorder.` (2)

All date values are converted to monthly datetime format, and all Google Trends variables are converted into numeric floating-point values. The six standardized country files are then concatenated into a single country-panel dataset.

2.3 Raw Panel Validation

Before cleaning, the raw monthly structure is validated. The expected start date is January 2004, and the expected end date is December 2025. Each country should therefore contain exactly 264 monthly observations with no missing months and no duplicated monthly timestamps.

The validation confirms that all six countries satisfy the expected monthly structure. No duplicated months are detected, and no monthly periods are missing from the raw panel. This means that the raw data form a balanced panel before zero-value treatment and interpolation.

Table 3: Expected Raw Panel Structure

Country	Start	End	Months	Structure check
Australia	2004-01	2025-12	264	Pass
Canada	2004-01	2025-12	264	Pass
Ireland	2004-01	2025-12	264	Pass
New Zealand	2004-01	2025-12	264	Pass
UK	2004-01	2025-12	264	Pass
US	2004-01	2025-12	264	Pass

2.4 Treatment of Google Trends Zero Values

Google Trends zero values are treated cautiously. A value of zero in Google Trends does not necessarily mean that search interest was literally zero. It may also indicate insufficient search volume relative to the Google Trends scaling procedure. Therefore,

zeros are treated as potentially low-volume observations rather than as true zero search interest.

The preprocessing rule is:

1. convert zero values in Google Trends variables to missing values;
2. apply conservative short-gap linear interpolation;
3. keep track of all originally zero or missing observations using imputation flags;
4. avoid over-imputing long low-volume gaps;
5. create a complete modeling file only after transparent flags have been added.

The raw data contain no originally missing values, but several zero values are present in some country-variable pairs. The zero-value summary is shown in Table 4.

Table 4: Raw Zero-Value Summary

Country	Variable	Raw zero count
Ireland	Anxiety	4
Ireland	Stress	3
Ireland	Insomnia	26
Ireland	Mental disorder	4
New Zealand	Stress	3
New Zealand	Insomnia	22
UK	Stress	2

Aggregated by variable, the number of values converted from zero to missing is 4 for anxiety, 0 for depression, 8 for stress, 48 for insomnia, and 4 for mental disorder. Insomnia contains the highest number of low-volume zero observations, especially in Ireland and New Zealand.

2.5 Short-Gap Linear Interpolation

After converting zeros to missing values, short gaps are filled using linear interpolation within each country. The interpolation rule is intentionally conservative:

```
interpolate(method = ‘‘linear’’, limit = 3, limit_direction = ‘‘both’’).  
(3)
```

This means that short internal gaps are filled linearly, and short leading or trailing gaps of up to three months can be filled from the nearest observed values. Longer leading gaps are not fully interpolated because doing so would risk artificially creating early-period search behavior where Google Trends records very low search volume.

After short-gap interpolation, all missing values are removed for anxiety, depression, stress, and mental disorder. The only remaining missing observations are six insomnia

values for New Zealand from January 2004 to June 2004. These observations remain missing in the transparent EDA file but are later filled in the modeling-ready file with imputation flags.

Table 5: Missing Values Before and After Short-Gap Interpolation

Variable	After zero-to-missing conversion	After interpolation
Anxiety	4	0
Depression	0	0
Stress	8	0
Insomnia	48	6
Mental disorder	4	0

2.6 Clean EDA Dataset

The first cleaned output is `input.csv`. This file is designed for transparent exploratory data analysis. It preserves the cleaned country-panel structure but allows a small number of remaining missing values if they represent long leading low-volume periods that should not be aggressively imputed.

The resulting EDA dataset contains 1,584 rows and 7 columns:

`date, country, anxiety, depression, stress, insomnia, mental_disorder.` (4)

The file contains six remaining missing values, all corresponding to early New Zealand insomnia observations. This distinction is important because the EDA dataset prioritizes transparency, while the modeling dataset requires complete monthly series.

2.7 Modeling-Ready Dataset

The second cleaned output is `input_modeling.csv`. This file is used for ARIMA, SARIMA, and SARIMAX modeling. Since these models require complete monthly series, the remaining edge missing values after interpolation are filled within each country using forward and backward filling.

Before this final filling step, variable-specific imputation flags are created. For each mental-health variable, a flag equals one if the original raw value was zero or missing, or if the value remained missing after conservative interpolation. This design keeps all imputed values traceable during modeling.

The modeling-ready dataset contains 1,584 rows and 20 columns. It has no missing values. The retained modeling variables are the five Google Trends variables, five imputation flags, calendar features, time index, phase labels, and COVID intervention variables.

Table 6: Preprocessing Output Files

File	Shape	Purpose
<code>input.csv</code>	$1,584 \times 7$	Clean EDA dataset
<code>input_modeling.csv</code>	$1,584 \times 20$	Complete ARIMA/SARIMA/SARIMAX dataset
<code>input_modeling_long.csv</code>	$7,920 \times 13$	Long-format plotting and comparison dataset
<code>modeling_data_dictionary.md</code>	–	Variable and preprocessing documentation

2.8 COVID Intervention Variables

The modeling-ready dataset includes intervention variables designed for SARIMAX analysis. The sample is divided into three analysis phases: pre-COVID, COVID-period, and post-COVID. The pre-COVID period runs from January 2004 to December 2019. The COVID-period intervention window runs from January 2020 to May 2023. The post-COVID period starts in June 2023 and ends in December 2025.

Table 7: COVID Phase Definition

Phase	Period	Months per country	Interpretation
Pre-COVID	2004-01 to 2019-12	192	Baseline period
COVID period	2020-01 to 2023-05	41	Intervention period
Post-COVID	2023-06 to 2025-12	31	Post-intervention period

The SARIMAX intervention variables are defined as follows:

- `covid_period`: dummy equal to 1 from January 2020 to May 2023, and 0 otherwise;
- `covid_trend`: monthly counter inside the COVID period, running from 1 to 41 and equal to 0 outside the COVID period;
- `post_covid_period`: dummy equal to 1 from June 2023 onward, and 0 otherwise;
- `post_covid_trend`: monthly counter inside the post-COVID period, running from 1 to 31 and equal to 0 before June 2023.

The level dummies measure average shifts during and after the COVID period, while the trend variables measure whether the effect strengthens, fades, or continues over time.

2.9 Long-Format Dataset

For plotting, heatmaps, and cross-variable comparisons, the wide modeling file is also converted into long format. The long-format file is saved as `input_modeling_long.csv`. It contains 7,920 rows, corresponding to:

$$1,584 \text{ country-month observations} \times 5 \text{ mental-health variables} = 7,920 \text{ rows.} \quad (5)$$

Each row contains the date, country, variable name, value, imputation flag, calendar information, analysis phase, and COVID intervention variables. This format is mainly used for EDA visualizations and cross-country comparisons.

2.10 Connection to the Modeling Stage

The preprocessing design is aligned with the later ARIMA, SARIMA, and SARIMAX modeling pipeline. The final modeling file provides complete monthly series for each country, a consistent monthly frequency, interpretable intervention variables, and traceable imputation flags.

The modeling stage uses the following time split:

Table 8: Modeling Split Design

Split	Period	Purpose
Training set	2004-01 to 2021-12	Estimate candidate models
Validation set	2022-01 to 2023-05	Select ARIMA/SARIMA/SARIMAX specification
Test set	2023-06 to 2025-12	Final out-of-sample forecast evaluation
Pre-COVID fit	2004-01 to 2019-12	Estimate no-COVID counterfactual baseline
Counterfactual forecast	2020-01 to 2025-12	Compare observed data with no-COVID forecast

This split ensures that model selection and final forecast evaluation are chronologically valid. The test period corresponds to the post-COVID phase, which allows the study to evaluate whether mental-health search behavior returns to pre-COVID dynamics or remains structurally different after the main COVID intervention period.

2.11 Preprocessing Summary

The preprocessing stage transforms six raw Google Trends country files into a balanced and modeling-ready monthly panel. The raw data cover six countries from January 2004 to December 2025, producing 1,584 country-month observations. The raw panel passes the monthly structure validation with no missing or duplicated months.

Google Trends zero values are treated as low-volume or insufficient-volume observations rather than literal zero search interest. These values are first converted to missing values and then handled using conservative short-gap linear interpolation. A transparent EDA dataset is saved as `input.csv`, while a complete modeling dataset is saved as `input_modeling.csv`. The modeling dataset contains no missing values and includes imputation flags for all replaced observations.

Finally, COVID intervention variables are added to support SARIMAX intervention analysis. The data are divided into pre-COVID, COVID-period, and post-COVID phases, with 192, 41, and 31 months per country respectively. These preprocessing outputs provide the foundation for the EDA, ARIMA/SARIMA/SARIMAX modeling, forecast evaluation, and structural-break analysis in the following sections.

3 Exploratory Data Analysis

3.1 EDA Objective

This section examines the main time-series properties of anxiety-related Google Trends search interest before formal ARIMA, SARIMA, and SARIMAX modeling. The objective is not to select the final model visually, but to identify trend behavior, seasonality, stationarity, autocorrelation, and possible COVID-period structural breaks. The main variable is **anxiety**, while other mental-health search variables are used for descriptive context.

3.2 Long-Term Trend in Anxiety Search Interest

Anxiety search interest increased substantially across all six countries between 2004 and 2025. Search levels were relatively low and stable in the early sample, then rose more clearly after the early 2010s. During and after the COVID period, anxiety search interest remained high, although the pattern became more volatile and country-specific.

This upward movement suggests that the level series is unlikely to be stationary. Therefore, first differencing is required before ARIMA and SARIMA modeling. The strong post-2020 levels also motivate SARIMAX intervention terms to distinguish COVID-period changes from pre-existing trend behavior.

3.3 COVID-Period and Post-COVID Phase Comparison

Mean anxiety search interest is much higher in the COVID and post-COVID periods than in the pre-COVID period across all countries. This indicates that anxiety-related search behavior did not return to its earlier pre-COVID level after the main COVID window ended.

Table 9: Mean Anxiety Search Interest by Phase

Country	Pre-COVID	COVID	Post-COVID	Post vs. Pre (%)
Australia	49.59	86.71	84.06	69.52
Canada	49.68	88.83	84.65	70.37
Ireland	44.00	82.37	76.97	74.92
New Zealand	47.31	89.07	84.06	77.70
UK	40.00	88.56	86.84	117.10
US	45.89	85.37	90.16	96.47

However, this phase comparison is descriptive only. It does not control for trend, seasonality, or autocorrelation. Therefore, it cannot be interpreted as causal evidence that COVID directly increased anxiety searches. Formal SARIMAX intervention analysis is required for that purpose.

3.4 Seasonality

Monthly averages show mild but visible seasonality in anxiety search interest. The peak months differ across countries, but recurring month-of-year patterns are present. This supports the use of SARIMA models with monthly seasonal period $s = 12$.

The seasonal coefficient of variation for anxiety is lower than for several other mental-health variables, meaning that anxiety has seasonality but is more strongly driven by trend and persistence. Therefore, SARIMA terms are justified, but SARIMA is not guaranteed to outperform simple seasonal benchmarks.

3.5 Cross-Country and Within-Country Patterns

Anxiety search trends are highly synchronized across countries after z-score normalization, with pairwise correlations above 0.95. This suggests that anxiety-related search behavior follows a common international pattern rather than being purely country-specific.

Within each country, anxiety is generally positively correlated with insomnia and more weakly related to the other mental-health search variables. Depression and stress do not follow the same stable upward pattern as anxiety. This supports the decision to use anxiety as the main modeling variable.

3.6 Stationarity Tests

Stationarity is evaluated using the Augmented Dickey–Fuller test and the KPSS test. The level anxiety series is non-stationary in all six countries. After first differencing, the series becomes much closer to stationary. This supports using $d = 1$ in the ARIMA-family models.

Table 10: Stationarity Tests for Anxiety Series

Country	ADF p-level	KPSS p-level	ADF p-diff	KPSS p-diff
Australia	0.733	0.010	0.046	0.100
Canada	0.701	0.010	0.053	0.100
Ireland	0.727	0.010	0.011	0.100
New Zealand	0.765	0.010	0.000	0.100
UK	0.821	0.010	0.011	0.100
US	0.942	0.010	0.000	0.100

3.7 ACF and PACF Analysis

The ACF and PACF of first-differenced anxiety show two main patterns. First, most countries display short-run dependence around lag 1, which supports low-order ARIMA candidates such as ARIMA(0, 1, 1), ARIMA(1, 1, 1), ARIMA(2, 1, 1), and ARIMA(1, 1, 2). Second, the ACF shows seasonal spikes at lags 12, 24, and 36, supporting SARIMA and SARIMAX models with seasonal period $s = 12$.

Table 11: EDA-Based Candidate Model Logic

EDA Finding	Modeling Implication
Level series is non-stationary	Use $d = 1$
Lag-1 dependence	Test ARIMA(0, 1, 1) and ARIMA(1, 1, 1)
Lag 1–2 dependence in some countries	Test ARIMA(2, 1, 1) and ARIMA(1, 1, 2)
Seasonal spikes at 12, 24, and 36	Use seasonal period $s = 12$
Uncertain seasonal differencing strength	Compare $D = 0$ and $D = 1$
Seasonal AR and MA behavior both plausible	Test seasonal AR and seasonal MA terms

3.8 Structural-Break Motivation

Structural-break tests indicate significant changes around both the COVID onset and the post-COVID transition in all six countries. These tests include a time trend and monthly seasonal controls, so the result is not based only on raw mean differences.

Table 12: Structural Break Tests for Anxiety

Country	Break Date	F-statistic	p-value	Significant
Australia	2020-01	10.857	0.000	Yes
Canada	2020-01	9.955	0.000	Yes
Ireland	2020-01	10.724	0.000	Yes
New Zealand	2020-01	7.918	0.000	Yes
UK	2020-01	6.595	0.000	Yes
US	2020-01	2.258	0.008	Yes
Australia	2023-06	8.031	0.000	Yes
Canada	2023-06	8.468	0.000	Yes
Ireland	2023-06	7.743	0.000	Yes
New Zealand	2023-06	6.635	0.000	Yes
UK	2023-06	5.063	0.000	Yes
US	2023-06	1.946	0.026	Yes

These results motivate SARIMAX intervention variables for the COVID and post-COVID periods. However, the break tests do not prove causality; they only show that the time-series relationship changes around the selected break dates.

3.9 EDA Summary

The EDA supports the modeling strategy used in the next sections. Anxiety search interest increased strongly across countries, the level series are non-stationary, first differencing is appropriate, and ACF/PACF patterns justify low-order ARIMA models with monthly SARIMA components. Structural-break tests further motivate SARIMAX intervention models. Therefore, ARIMA is used to model non-seasonal persistence, SARIMA is used to model annual monthly seasonality, and SARIMAX is used to estimate COVID-period and post-COVID intervention effects.

4 Modeling

4.1 Modeling Design

This section estimates ARIMA, SARIMA, and SARIMAX models for monthly anxiety search interest across six countries. The purpose is twofold. First, the models are used for out-of-sample forecasting. Second, SARIMAX is used as an intervention model to examine whether anxiety search behavior changed during and after the COVID period.

The modeling sample is divided chronologically into training, validation, and test sets. Random splitting is not used because it would violate the time-series structure.

Table 13: Forecasting Split Design

Split	Period	Purpose
Training	2004-01 to 2021-12	Estimate candidate models
Validation	2022-01 to 2023-05	Select best model specification
Test	2023-06 to 2025-12	Final out-of-sample evaluation

The test period corresponds to the post-COVID phase. This allows the final forecast evaluation to test whether models estimated from pre-COVID and COVID data can predict post-COVID anxiety search behavior.

4.2 Benchmark Models

Two simple benchmark models are included: Naive and Seasonal Naive. The Naive model uses the most recent observed value as the forecast:

$$\hat{y}_{t+h} = y_t. \quad (6)$$

The Seasonal Naive model uses the value from the same month in the previous year:

$$\hat{y}_{t+h} = y_{t+h-12}. \quad (7)$$

These benchmarks are necessary because anxiety search interest is highly persistent and seasonal. ARIMA-family models are only useful for forecasting if they improve upon these simple alternatives.

4.3 ARIMA and SARIMA Models

The ARIMA model captures non-seasonal time-series dependence after differencing:

$$\phi(B)(1-B)^d y_t = c + \theta(B)\varepsilon_t. \quad (8)$$

Based on the stationarity tests in the EDA section, the candidate ARIMA models use first differencing, so $d = 1$. The tested ARIMA orders are:

$$(0, 1, 1), (1, 1, 1), (2, 1, 1), (1, 1, 2), (2, 1, 2), (3, 1, 1), (1, 1, 3). \quad (9)$$

Because the EDA shows seasonal spikes at lags 12, 24, and 36, SARIMA models are also estimated using monthly seasonality, with $s = 12$:

$$\Phi(B^{12})\phi(B)(1-B)^d(1-B^{12})^D y_t = c + \Theta(B^{12})\theta(B)\varepsilon_t. \quad (10)$$

The SARIMA candidate set compares both $D = 0$ and $D = 1$, as well as seasonal AR and seasonal MA terms. The candidate grid is deliberately moderate to avoid overfitting a 264-month series.

4.4 SARIMAX Intervention Model

SARIMAX extends SARIMA by adding intervention variables:

$$y_t = \beta_0 + \beta_1 \text{early_covid_shock}_t + \beta_2 \text{covid_period}_t + \beta_3 \text{covid_trend}_t + n_t, \quad (11)$$

where n_t follows a SARIMA error process.

For forecasting, only the intervention variables observable before the test period are used: `early_covid_shock`, `covid_period`, and `covid_trend`. The post-COVID variables are not used in forecast model selection because they cannot be learned before the post-COVID test period.

For full-sample explanatory analysis, the SARIMAX model additionally includes:

$$\text{post_covid_period}_t \text{ and } \text{post_covid_trend}_t. \quad (12)$$

This full-sample SARIMAX specification is interpreted as an intervention model, not as causal proof of clinical mental-health changes.

4.5 Model Selection

Candidate models are first estimated on the training sample and evaluated on the validation sample. The best model within each class is selected using validation RMSE, with AIC and BIC used only as secondary criteria.

Table 14: Selected ARIMA, SARIMA, and SARIMAX Specifications by Validation RMSE

Country	ARIMA	SARIMA	SARIMAX
Australia	(3, 1, 1)	(1, 1, 1)(1, 1, 1) ₁₂	(1, 1, 1)(0, 1, 1) ₁₂
Canada	(2, 1, 1)	(1, 1, 1)(0, 0, 1) ₁₂	(1, 1, 1)(0, 0, 1) ₁₂
Ireland	(0, 1, 1)	(2, 1, 1)(0, 1, 1) ₁₂	(2, 1, 1)(0, 1, 1) ₁₂
New Zealand	(2, 1, 2)	(0, 1, 1)(1, 0, 0) ₁₂	(0, 1, 1)(1, 0, 0) ₁₂
UK	(0, 1, 1)	(0, 1, 1)(0, 0, 1) ₁₂	(2, 1, 1)(0, 0, 1) ₁₂
US	(3, 1, 1)	(1, 1, 1)(1, 1, 1) ₁₂	(1, 1, 1)(1, 1, 1) ₁₂

The selected specifications confirm the EDA findings. All selected ARIMA-family models use first differencing, which is consistent with the non-stationarity of the level series. Most SARIMA and SARIMAX models also include seasonal terms, supporting the presence of annual monthly seasonality.

4.6 Validation Results

The validation results show that the best model differs by country. SARIMA is best for Australia, Naive is best for Canada, Ireland, and the UK, SARIMAX is best for New Zealand and the US. This means that no single model class dominates during the validation period.

Table 15: Best Validation Model by Country

Country	Best validation model	RMSE	sMAPE
Australia	SARIMA	4.917	4.840
Canada	Naive	6.494	5.978
Ireland	Naive	5.504	5.660
New Zealand	SARIMAX	6.196	5.324
UK	Naive	7.034	6.858
US	SARIMAX	4.263	4.186

The validation stage therefore supports keeping all model classes for the final test evaluation. ARIMA, SARIMA, and SARIMAX remain relevant, but benchmark models are also competitive and cannot be ignored.

4.7 Residual Diagnostics

Residual diagnostics are used to check whether the selected models leave systematic autocorrelation. The diagnostics include Ljung–Box tests at lags 12 and 24, ARCH-LM tests, and Jarque–Bera normality tests.

The diagnostic results show that SARIMA and SARIMAX generally reduce residual autocorrelation more effectively than non-seasonal ARIMA. However, some residual autocorrelation and heteroskedasticity remain. This is expected because Google Trends anxiety search behavior contains strong structural changes and post-COVID instability. The residual diagnostics therefore support a cautious interpretation of forecast results rather than claiming that any model fully captures the data-generating process.

4.8 Modeling Summary

The modeling stage estimates benchmark, ARIMA, SARIMA, and SARIMAX models using a consistent country-level pipeline. ARIMA captures non-seasonal persistence, SARIMA adds annual monthly seasonality, and SARIMAX adds COVID-related intervention variables. Model selection is based on validation RMSE rather than in-sample fit alone.

The main conclusion from this stage is that the data require both differencing and seasonal modeling, but greater model complexity does not guarantee better forecast performance. The final test-period comparison is therefore reported separately in the forecast evaluation section.

5 Forecast Evaluation

5.1 Evaluation Design

Forecast evaluation is conducted on the post-COVID test period from June 2023 to December 2025. This period contains 31 monthly observations for each country. All models are compared using the same test window to ensure a fair out-of-sample comparison.

The evaluated models are Naive, Seasonal Naive, ARIMA, SARIMA, and SARIMAX. Forecast accuracy is measured using MAE, RMSE, and sMAPE. MAE measures average absolute error, RMSE penalizes large errors more strongly, and sMAPE reports scale-adjusted percentage error.

5.2 Average Test Performance

Table 16 reports average test performance across all six countries.

Table 16: Average Test Performance by Model Class

Model	MAE	RMSE	sMAPE
Seasonal Naive	4.715	5.989	5.543
Naive	7.876	9.312	9.060
ARIMA	8.745	10.688	10.093
SARIMA	9.565	11.395	10.823
SARIMAX	21.883	23.169	22.751

The Seasonal Naive model has the lowest average error across all three metrics. This shows that anxiety search interest in the post-COVID period is strongly seasonal and persistent. Simply using the value from the same month in the previous year gives more accurate forecasts than the estimated ARIMA-family models over the full test horizon.

ARIMA and SARIMA do not outperform the benchmark on average. SARIMAX performs worst as a pure forecasting model. This does not make SARIMAX useless; it means that its main contribution in this project is intervention interpretation rather than post-COVID forecast accuracy.

5.3 Best Test Model by Country

Table 17 reports the best test-period model for each country.

Table 17: Best Test Forecast Model by Country

Country	Best model	RMSE	sMAPE
Australia	Seasonal Naive	5.221	4.512
Canada	Seasonal Naive	5.064	4.942
Ireland	Seasonal Naive	5.510	5.252
New Zealand	Seasonal Naive	8.227	7.402
UK	Seasonal Naive	4.711	4.422
US	Naive	4.299	3.510

Seasonal Naive is the best model in five out of six countries. The only exception is the United States, where the Naive model performs best. This result confirms that simple benchmark models dominate the full post-COVID test period. The result also suggests that the post-COVID anxiety series is difficult to forecast using estimated parametric dynamics.

5.4 Horizon-Specific Performance

The full 31-month test period is further divided into forecast horizons: 1–3 months, 4–12 months, 13–24 months, and 25 months or longer. This shows whether model performance changes with the forecast horizon.

Table 18: Horizon-Specific Average RMSE

Model	h01–03	h04–12	h13–24	h25+
Seasonal Naive	4.223	5.442	4.831	8.426
Naive	5.199	8.198	8.739	12.183
ARIMA	4.680	7.055	9.806	16.176
SARIMA	3.776	7.352	10.409	17.549
SARIMAX	15.112	16.970	23.048	31.294

SARIMA performs best at the very short horizon of one to three months, with an RMSE of 3.776. This indicates that seasonal ARIMA dynamics still contain useful short-term forecasting information. However, from horizon 4 onward, Seasonal Naive becomes the strongest model. This means that SARIMA has short-horizon value, but its advantage does not persist over longer post-COVID forecasts.

5.5 Rolling-Origin Evaluation

A rolling-origin evaluation is also used as a robustness check. In this design, the forecast origin is updated repeatedly over time rather than fixed once at the end of the training period.

Table 19: Rolling-Origin Average Performance

Model	MAE	RMSE	sMAPE
Seasonal Naive	4.767	5.730	5.539
SARIMA	5.481	6.528	6.345
ARIMA	5.554	6.881	6.489
Naive	9.333	10.470	11.333
SARIMAX	9.583	10.559	10.538

The rolling-origin results again rank Seasonal Naive first. SARIMA and ARIMA are closer under this evaluation design, but they still do not outperform Seasonal Naive. This confirms that benchmark dominance is not caused by only one fixed forecast split.

5.6 Diebold–Mariano Test

Diebold–Mariano tests are used to compare ARIMA-family models against the best country-level benchmark. A positive mean loss difference indicates that the ARIMA-family model has higher squared forecast loss than the benchmark.

Table 20: Diebold–Mariano Test Summary

Model	Tests	Significant at 5%	Mean loss difference
ARIMA	6	5	89.141
SARIMA	6	5	106.565
SARIMAX	6	5	601.331

The Diebold–Mariano results show that benchmark superiority is usually statistically meaningful. For ARIMA, SARIMA, and SARIMAX, five out of six tests are significant at the 5% level. The positive mean loss differences indicate that these models generally underperform the benchmark during the test period.

5.7 Forecast Evaluation Summary

The forecast evaluation produces four main findings. First, benchmark models dominate the full post-COVID test horizon. Seasonal Naive is best in five countries, while Naive is best in the United States. Second, SARIMA is useful at the short horizon of one to three months, where it achieves the lowest average RMSE. Third, the rolling-origin results confirm that Seasonal Naive remains the most robust forecasting method. Fourth, Diebold–Mariano tests show that the benchmark advantage is usually statistically significant.

Therefore, the final forecasting conclusion is conservative. ARIMA, SARIMA, and SARIMAX are useful for modeling time-series structure, seasonality, and intervention effects, but they do not provide the best full-horizon post-COVID forecasts. The strongest pure forecasting model in this test design is the Seasonal Naive benchmark.

6 Structural Break SARIMAX

6.1 Purpose

The forecast evaluation shows that benchmark models outperform ARIMA-family models for post-COVID forecasting. Therefore, SARIMAX is used mainly as an intervention model rather than as the best pure forecasting model. The purpose of this section is to examine whether anxiety search interest changed systematically during the COVID and post-COVID periods after controlling for trend, seasonality, and ARIMA-type serial dependence.

6.2 SARIMAX Intervention Specification

The full-sample SARIMAX model includes five intervention variables:

$$y_t = \beta_0 + \beta_1 \text{early_covid_shock}_t + \beta_2 \text{covid_period}_t + \beta_3 \text{covid_trend}_t + \beta_4 \text{post_covid_period}_t + \beta_5 \text{post_covid_trend}_t \quad (13)$$

where n_t follows the country-specific SARIMA error process selected in the modeling stage. The variable `early_covid_shock` captures the initial COVID shock, `covid_period` captures the average COVID-period level shift, and `covid_trend` captures the within-COVID slope change. The two post-COVID variables capture the level and trend behavior after June 2023.

6.3 Significant Intervention Terms

Table 21 reports the statistically significant SARIMAX intervention terms at the 5% level.

Table 21: Significant SARIMAX Intervention Terms

Country	Term	Coefficient	p-value
Australia	<code>covid_trend</code>	-0.519	0.030
Australia	<code>post_covid_period</code>	-23.760	0.025
Australia	<code>post_covid_trend</code>	-0.962	0.005
Canada	<code>post_covid_trend</code>	-0.647	0.002
UK	<code>early_covid_shock</code>	-9.869	0.000
UK	<code>post_covid_trend</code>	-0.526	0.035
US	<code>post_covid_trend</code>	-0.497	0.049

The results identify seven statistically significant intervention terms across four countries. All significant coefficients are negative. This is the key finding of the structural-break SARIMAX analysis: after controlling for the strong pre-existing upward trend, seasonal behavior, and serial dependence, the intervention evidence does not support a uniform positive COVID-period increase in anxiety search interest.

The strongest post-COVID decline is found in Australia, where both the post-COVID level term and post-COVID trend term are statistically significant. Canada, the UK, and the US also show significant negative post-COVID trend terms. Ireland and New Zealand do not have statistically significant intervention terms at the 5% level.

6.4 Coefficient Interpretation

The significant negative coefficients do not mean that anxiety search interest was low during COVID in raw levels. The descriptive EDA already showed that raw COVID and post-COVID anxiety levels were much higher than pre-COVID levels. Instead, the SARIMAX coefficients mean that observed anxiety search interest was lower than what would be expected after accounting for the strong historical upward trend and seasonal structure.

This distinction is important. The raw data show high anxiety search interest during and after COVID, but the intervention model asks a stricter question: whether COVID-period observations were unusually high relative to a model-based counterfactual trend. Under this specification, the answer is generally no.

6.5 Counterfactual Comparison

Counterfactual forecasts are estimated using pre-COVID dynamics and compared with observed anxiety search interest during the COVID and post-COVID periods. Because Google Trends indices are bounded between 0 and 100, the raw counterfactual version can exceed the valid scale. The raw version exceeds 100 in 49.1% of full-period observations, so the capped and logit versions are more appropriate for interpretation.

Table 22 reports the logit counterfactual summary.

Table 22: Logit Counterfactual Summary

Phase	Mean gap	Mean % gap	Share above counterfactual
COVID period	-2.848	-3.063	0.386
Full 2020–2025	-6.063	-6.459	0.252
Post-COVID	-10.315	-10.719	0.075

The logit counterfactual results show that observed anxiety search interest is, on average, below the counterfactual benchmark. During the COVID period, the average gap is -2.848 index points. Across the full 2020–2025 period, the average gap is -6.063 points. In the post-COVID period, the gap becomes larger, at -10.315 points. This supports the SARIMAX coefficient results: anxiety search interest remained high in absolute terms, but it did not exceed the level predicted by the strong pre-COVID upward trajectory.

6.6 Country-Level Counterfactual Results

The country-level logit counterfactual comparison shows heterogeneous patterns. During the COVID period, Canada and the US have small positive gaps, while Australia, Ireland, New Zealand, and the UK have negative gaps. In the post-COVID period, all countries have negative gaps.

Table 23: Post-COVID Logit Counterfactual Results by Country

Country	Observed mean	Counterfactual mean	Mean gap
Australia	84.065	95.861	-11.797
Canada	84.645	91.907	-7.262
Ireland	76.968	97.844	-20.876
New Zealand	84.065	97.739	-13.674
UK	86.839	94.595	-7.756
US	90.161	90.686	-0.525

Ireland and New Zealand have the largest negative post-COVID gaps, while the US has the smallest negative gap. This indicates that the post-COVID behavior is not identical across countries, even though all countries share a broadly similar long-term anxiety trend.

6.7 Residual Diagnostics

Residual diagnostics are used to check whether the full-sample SARIMAX models adequately remove serial correlation. The Ljung–Box test at lag 24 is passed in four out of six countries. Australia, Ireland, New Zealand, and the US pass the lag-24 residual autocorrelation test, while Canada and the UK do not.

Table 24: Full-Sample SARIMAX Residual Diagnostics

Country	Residual std.	Ljung–Box p(24)	Pass p(24) at 5%
Australia	4.184	0.101	Yes
Canada	4.377	0.000	No
Ireland	4.694	0.851	Yes
New Zealand	5.678	0.147	Yes
UK	3.698	0.000	No
US	3.441	0.143	Yes

The residual diagnostics imply that the intervention results should be interpreted cautiously, especially for Canada and the UK. The SARIMAX models capture part of the intervention structure, but they do not fully eliminate all remaining time-series dependence in every country.

6.8 Summary

The structural-break SARIMAX results provide a more cautious interpretation than the raw EDA. Raw anxiety search interest increased sharply during and after the COVID period, but the intervention models do not find robust positive COVID effects after controlling for long-term trend, seasonality, and serial dependence. Instead, all statistically significant intervention terms are negative or declining.

The main implication is that COVID-period anxiety search interest was high mainly because it occurred after a long pre-existing upward trend. The evidence does not support the stronger claim that COVID created a uniform additional positive shock to anxiety searches across all countries. SARIMAX is therefore most useful in this project as an intervention-interpretation tool, not as the best post-COVID forecasting model.

7 Cross-National Comparison

7.1 Purpose

This section compares the six countries across four dimensions: forecasting performance, SARIMAX intervention evidence, counterfactual gaps, and residual diagnostics. The purpose is to determine whether the empirical findings are common across countries or driven by a small number of country-specific cases.

7.2 Country Scorecard

Table 25 summarizes the main country-level results.

Table 25: Cross-National Country Scorecard

Country	Best test model	RMSE	sMAPE	Logit gap	Main interpretation
Australia	Seasonal Naive	5.221	4.512	-7.982	Benchmark-dominant; negative intervention evidence
Canada	Seasonal Naive	5.064	4.942	-2.949	Benchmark-dominant; negative post-COVID trend
Ireland	Seasonal Naive	5.510	5.252	-14.474	Benchmark-dominant; no significant intervention terms
New Zealand	Seasonal Naive	8.227	7.402	-8.031	Benchmark-dominant; no significant intervention terms
UK	Seasonal Naive	4.711	4.422	-3.655	Benchmark-dominant; negative intervention evidence
US	Naive	4.299	3.510	0.714	Benchmark-dominant; mildly positive counterfactual gap

The country scorecard shows that benchmark models dominate the post-COVID test period in all six countries. Seasonal Naive is the best model in five countries, while Naive is best in the United States. This confirms that the main forecasting result is not country-specific. Post-COVID anxiety search interest is better predicted by simple persistence and seasonal repetition than by fitted ARIMA-family models.

7.3 Forecasting Comparison

The cross-country forecasting results show a clear pattern. In Australia, Canada, Ireland, New Zealand, and the UK, the best model is Seasonal Naive. In the US, the best model is Naive. The implication is that the post-COVID anxiety series is highly

persistent. Recent values and same-month values from the previous year provide stronger forecast information than estimated ARIMA, SARIMA, or SARIMAX dynamics.

This does not eliminate the value of ARIMA-family models. SARIMA remains useful for short-horizon dynamics, especially at horizons of one to three months. However, for full post-COVID test-period forecasting, the benchmark models are more robust.

7.4 Intervention Evidence

SARIMAX intervention results are heterogeneous across countries. Significant intervention terms are found in four countries: Australia, Canada, the UK, and the US. Ireland and New Zealand do not have statistically significant SARIMAX intervention terms at the 5% level.

All significant SARIMAX intervention coefficients are negative or declining. Australia shows significant negative coefficients for `covid_trend`, `post_covid_period`, and `post_covid_trend`. Canada, the UK, and the US show significant negative post-COVID trend effects. The UK also has a significant negative `early_covid_shock` coefficient.

This means that the SARIMAX evidence does not support a robust positive COVID-period increase after controlling for trend, seasonality, and serial dependence. The raw anxiety index is high during and after COVID, but this high level is largely explained by the strong pre-existing upward trend.

7.5 Counterfactual Comparison

The logit counterfactual results show negative full-period mean gaps in five out of six countries. Australia, Canada, Ireland, New Zealand, and the UK have observed anxiety search interest below the bounded no-COVID counterfactual on average. The US is the only country with a mildly positive average gap.

The largest negative gap is found in Ireland, with a mean gap of -14.474 index points. New Zealand and Australia also show large negative gaps, at -8.031 and -7.982 respectively. The US has a small positive gap of 0.714, meaning that its observed anxiety search interest is slightly above the logit counterfactual on average.

These results reinforce the structural-break interpretation. The apparent COVID-period increase in raw anxiety search interest is not enough to establish an additional positive COVID intervention effect once the pre-existing trend is used as a counterfactual benchmark.

7.6 Residual Diagnostics

Residual diagnostics are mixed across countries. The Ljung–Box test at lag 24 is passed in four countries: Australia, Ireland, New Zealand, and the US. Canada and the UK fail the lag-24 residual autocorrelation test, meaning that their SARIMAX residuals still contain remaining serial dependence.

This matters for interpretation. The intervention results are more reliable in countries where residual autocorrelation is adequately controlled. For Canada and the UK, the significant intervention coefficients should be interpreted with more caution because the model does not fully remove serial correlation.

7.7 Summary

The cross-national comparison produces five conclusions. First, benchmark models are best in all six post-COVID test countries, with Seasonal Naive best in five. Second, SARIMA remains useful for short-horizon seasonal dynamics but does not dominate full-horizon forecasting. Third, SARIMAX is more valuable as an intervention model than as a forecasting model. Fourth, the intervention evidence does not support a uniform positive COVID effect; all significant coefficients are negative or declining. Fifth, the logit counterfactual shows negative average gaps in five out of six countries, with only the US showing a small positive gap.

Overall, the six-country comparison supports a conservative interpretation: anxiety search interest rose strongly in raw levels, but this increase was part of a longer pre-existing upward trajectory. After accounting for trend, seasonality, and serial dependence, the evidence for an additional positive COVID-period shock is weak.

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